

Query Analysis

Query 1: Display All Student Records

SQL Query

```
SELECT * FROM Students;
```

Result

Displays all student records stored in the **Students** table.

Explanation

This query retrieves every record from the Students table. It is mainly used to verify that all student information has been inserted correctly into the database.

	studentsid [PK] integer ↗	studentname character varying (100) ↗	gender character (1) ↗	age integer ↗	grade character varying (2) ↗	mathscore integer ↗	sciencscore integer ↗	englishscore integer ↗
1	1	Aarav Patel	M	18	A	95	91	89
2	2	Diya Shah	F	17	A	88	92	94
3	3	Vivaan Mehta	M	19	B	76	81	79
4	4	Ananya Joshi	F	18	A	91	87	93
5	5	Krish Desai	M	17	C	64	71	68
6	6	Riya Patel	F	18	B	83	85	88
7	7	Aditya Singh	M	19	B	79	82	75
8	8	Meera Nair	F	18	A	96	95	98
9	10	Sneha Kapoor	F	19	B	84	80	86
10	11	Yash Patel	M	18	A	90	89	87
11	12	Kavya Sharma	F	17	A	93	94	91
12	13	Arjun Rao	M	18	B	72	74	70
13	14	Ishita Gupta	F	19	A	89	90	92
14	15	Vaghela Miraj	M	18	A	92	94	90
15	9	Rahul Verma	M	20	A	58	63	61

Query 2: Average Score in Each Subject

SQL Query

```
SELECT
ROUND(AVG(MathScore),2) AS Avg_MathScore,
ROUND(AVG(ScienceScore),2) AS Avg_ScienceScore,
ROUND(AVG(EnglishScore),2) AS Avg_EnglishScore
FROM Students;
```

Result

Returns the average marks obtained by students in Mathematics, Science, and English.

Explanation

This query uses the **AVG()** aggregate function to calculate the average score for each subject. It helps evaluate the overall academic performance of the class.

	avg_mathscore numeric	avg_sciencescore numeric	avg_englishscore numeric
1	83.33	84.53	84.07

Query 3: Top Performer (Highest Total Score)

SQL Query

```
SELECT StudentName,
(MathScore + ScienceScore + EnglishScore) AS Total_Score
FROM Students
ORDER BY Total_Score DESC
LIMIT 1;
```

Result

Displays the student who achieved the highest total score.

Explanation

The total marks are calculated by adding the scores of all three subjects. The results are sorted in descending order, and the student with the highest total score is displayed.

	studentname character varying (100)	total_score integer
1	Meera Nair	289

Query 4: Count Students Per Grade

SQL Query

```
SELECT Grade,
COUNT(*) AS Total_Students
FROM Students
GROUP BY Grade;
```

Result

Displays the number of students in each grade.

Explanation

This query groups students according to their grade and counts the number of students in each group using the **COUNT()** function.

	grade character varying (2) 🔒	total_students bigint 🔒
1	B	5
2	C	1
3	A	9

Query 5: Average Score by Gender

SQL Query

```
SELECT Gender,
ROUND(AVG((MathScore + ScienceScore + EnglishScore)/3.0),2) AS Avg_Score
FROM Students
GROUP BY Gender
ORDER BY Avg_Score DESC;
```

Result

Displays the average score of male and female students.

Explanation

The query calculates the average marks obtained by each gender. It helps compare the academic performance between male and female students.

	gender character (1) 🔒	avg_score numeric 🔒
1	F	89.95
2	M	78.75

Query 6: Students with Math Score Greater Than 80

SQL Query

```
SELECT StudentName, MathScore
FROM Students
WHERE MathScore > 80
ORDER BY MathScore DESC;
```

Result

Displays all students who scored more than 80 marks in Mathematics.

Explanation

This query filters student records using the **WHERE** clause and displays only those students whose Mathematics score is greater than 80.

	studentname character varying (100) 🔒	mathscore integer 🔒
1	Meera Nair	96
2	Aarav Patel	95
3	Kavya Sharma	93
4	Vaghela Miraj	92
5	Ananya Joshi	91
6	Yash Patel	90
7	Ishita Gupta	89
8	Diya Shah	88
9	Sneha Kapoor	84
10	Riya Patel	83

Query 7: Update Student Grade

SQL Query

```
UPDATE Students
SET Grade='A'
WHERE StudentName='Rahul Verma';
```

Verification Query

```
SELECT StudentName, Grade
FROM Students
WHERE StudentName='Rahul Verma';
```

Result

The student's grade is updated successfully and verified.

Explanation

The **UPDATE** statement modifies an existing record in the database. The verification query confirms that the student's grade has been updated correctly.

	studentname character varying (100) 🔒	grade character varying (2) 🔒
1	Rahul Verma	A

Query 8: Highest Math Score

SQL Query

```
SELECT StudentName, MathScore
FROM Students
WHERE MathScore =
(
SELECT MAX(MathScore)
FROM Students
);
```

Result

Displays the student(s) who obtained the highest Mathematics score.

Explanation

The inner query finds the maximum Mathematics score using **MAX()**. The outer query retrieves the student details whose Mathematics score matches the maximum value.

	studentname character varying (100) 🔒	mathscore integer 🔒
1	Meera Nair	96

Query 9: Lowest Total Score

SQL Query

```
SELECT StudentName,
(MathScore + ScienceScore + EnglishScore) AS Total_Score
FROM Students
ORDER BY Total_Score ASC
LIMIT 1;
```

Result

Displays the student with the lowest total score.

Explanation

The total marks are calculated for each student, and the records are sorted in ascending order. The student with the lowest total score is displayed.

	studentname character varying (100) 🔒	total_score integer 🔒
1	Rahul Verma	182

Query 10: Students Above Average Math Score

SQL Query

```
SELECT StudentName, MathScore
FROM Students
WHERE MathScore >
(
SELECT AVG(MathScore)
FROM Students
);
```

Result

Displays students whose Mathematics score is greater than the average Mathematics score.

Explanation

The subquery calculates the average Mathematics score, and the outer query displays only those students whose Mathematics score is above that average.

	studentname character varying (100) 🔒	mathscore integer 🔒
1	Aarav Patel	95
2	Diya Shah	88
3	Ananya Joshi	91
4	Meera Nair	96
5	Sneha Kapoor	84
6	Yash Patel	90
7	Kavya Sharma	93
8	Ishita Gupta	89
9	Vaghela Miraj	92

Query 11: Average Age by Grade

SQL Query

```
SELECT Grade,
ROUND(AVG(Age),2) AS Avg_Age
FROM Students
GROUP BY Grade
ORDER BY Grade;
```

Result

Displays the average age of students in each grade.

Explanation

This query groups students by grade and calculates the average age of students in each group using

the **AVG()** function.

	grade character varying (2)	avg_age numeric
1	A	18.11
2	B	18.60
3	C	17.00

Query 12: Total Score and Percentage

SQL Query

```
SELECT StudentName,  
(MathScore + ScienceScore + EnglishScore) AS Total_Score,  
ROUND(((MathScore + ScienceScore + EnglishScore)/300.0)*100,2) AS Percentage  
FROM Students  
ORDER BY Total_Score DESC;
```

Result

Displays the total marks and percentage of every student.

Explanation

This query calculates the total marks by adding the scores of all three subjects and computes the percentage based on a maximum score of 300. The results are sorted from highest to lowest total score.

	studentname character varying (100)	total_score integer	percentage numeric
1	Meera Nair	289	96.33
2	Kavya Sharma	278	92.67
3	Vaghela Miraj	276	92.00
4	Aarav Patel	275	91.67
5	Diya Shah	274	91.33
6	Ishita Gupta	271	90.33
7	Ananya Joshi	271	90.33
8	Yash Patel	266	88.67
9	Riya Patel	256	85.33
10	Sneha Kapoor	250	83.33
11	Vivaan Mehta	236	78.67
12	Aditya Singh	236	78.67
13	Arjun Rao	216	72.00
14	Krish Desai	203	67.67
15	Rahul Verma	182	60.67